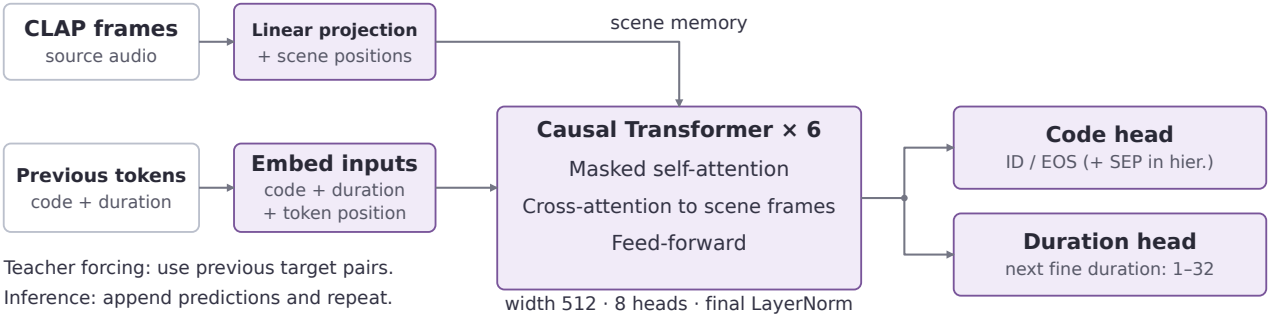
Inside the acoustic content LM

Learned / fine-tuned · Frozen · Fixed operation

(a) Prepare training targets from paired speech

Dashed arrows: target preparation only. Fit CTC with transcripts and the codebook with k-means, then freeze both.

(b) One shared Transformer, two prediction heads



Teacher forcing: use previous target pairs.
Inference: append predictions and repeat.

Training loss: code cross-entropy + fine-duration cross-entropy. The renderer is not optimized in this stage.

(c) Flat and hierarchical: different sequences, the same mechanism

Flat BOS → (c_1, d_1) → (c_2, d_2) → … → EOS

Hierarchical BOS → c_1 → c_{13} → … → SEP → (c_1, d_1) → (c_2, d_2) → … → EOS

Coarse prefix: every 12th fine token; no durations. Only the fine pairs are rendered.